

Arnav Tripathi

+91 9425963677 | arnavtripathi581@gmail.com | arnav-tripathi-4a95601a2/ | github.com/ArnavXT

EDUCATION

VIT Bhopal University

Bachelor's Degree | CGPA: 8.23

Bhopal, India

2023 – 2027

Macrovision Academy

CBSE 12th Board – 74.2%

Burhanpur, India

2023

Bhandari Public School

CBSE 10th Board – 92%

Khandwa, India

2021

EXPERIENCE

Software Development Intern – Backend Engineering

May 2026 – Present

SwarnShruti Technologies India Pvt. Ltd.

Remote

- Engineered a centralized FastAPI backend for Orchestra, a multi-platform developer activity aggregator, using serverless PostgreSQL and SQLAlchemy ORM
- Built real-time webhook ingestion pipelines (GitHub, Discord, Figma) with HMAC-SHA256 security and an async WebSocket-driven state machine for live task updates
- Implemented multi-provider OAuth (GitHub, Discord, Google) and secure backend proxy routes to protect API keys, enabling AI-powered roadmap generation and Neo4j graph syncing
- Collaborated with a 5-member distributed team to resolve schema mismatches and maintain zero-downtime integration

Virtual Intern – Google Cloud Generative AI

July 2025 – August 2025

NASSCOM

Remote

- Completed a virtual internship on Google Cloud Generative AI covering prompt engineering and AI model usage
- Gained hands-on experience with cloud platform deployment and real-world AI applications
- Leveraged Generative AI tools for innovation and problem-solving use cases

PROJECTS

Smart Traffic Violation Reporting System | [\[link\]](#)

2024

- Secured **Second Runner-up** at SOLVIT Hackathon competing against 150+ teams
- Built a system using advanced image recognition to detect violations (speeding, red-light running, no seat belt) in real time
- Implemented automated evidence capture (images/video) with instant alert dispatch to law enforcement agencies

Pathwise | Frontend Developer | [\[link\]](#)

Nov 2025 – Mar 2026

- Built an interactive React 19 + Vite UI with Framer Motion animations and Recharts for data visualization
- Integrated AI-generated course content with Markdown/LaTeX rendering, quizzes, and gamification features
- Implemented PDF/ZIP export functionality and OpenRouter AI integration for dynamic learning experiences

Project Aura: AI Chronic Disease Management

March 2026

- Developed an end-to-end AI platform for chronic disease management, integrating hybrid models (Random Forest & LSTM) for predictive health analytics
- Built a scalable **FastAPI** backend to handle real-time dataset ingestion and high-concurrency model inference requests
- Implemented specialized data preprocessing pipelines for multi-modal health records to ensure high prediction sensitivity

CERTIFICATIONS

NPTEL | Introduction to Machine Learning – IIT Madras

63% Score

NPTEL | Marketing Analytics – IIT Kharagpur

96% Score

Google | Technical IT Support Certificate

Oracle | Certified Generative AI Professional

Coursera | Bits and Bytes of Computer Networking

VITYarthi | AIML, Programming in Java, Python, and C++

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, Java

Frameworks & Libraries: FastAPI, React 19, SQLAlchemy, Recharts, Scikit-learn, TensorFlow/Keras, Pandas, NumPy

Backend & Systems: PostgreSQL, WebSockets, OAuth 2.0, REST APIs, Webhooks, APScheduler, Neo4j

AI/ML: Random Forest, LSTM, Computer Vision, Image Recognition, Generative AI, Prompt Engineering

Developer Tools: VS Code, Git/GitHub, Figma, Canva

Platforms & Cloud: AWS

Data & Analytics: Python, SQL/MySQL, Data Analysis, Machine Learning, Generative AI, Prompt Engineering, Statistical Modelling

EXTRA-CURRICULAR ACTIVITIES

Design Team Co-Lead

WiCyS VIT Bhopal Student Chapter

Design Team Member

E-Cell VIT Bhopal